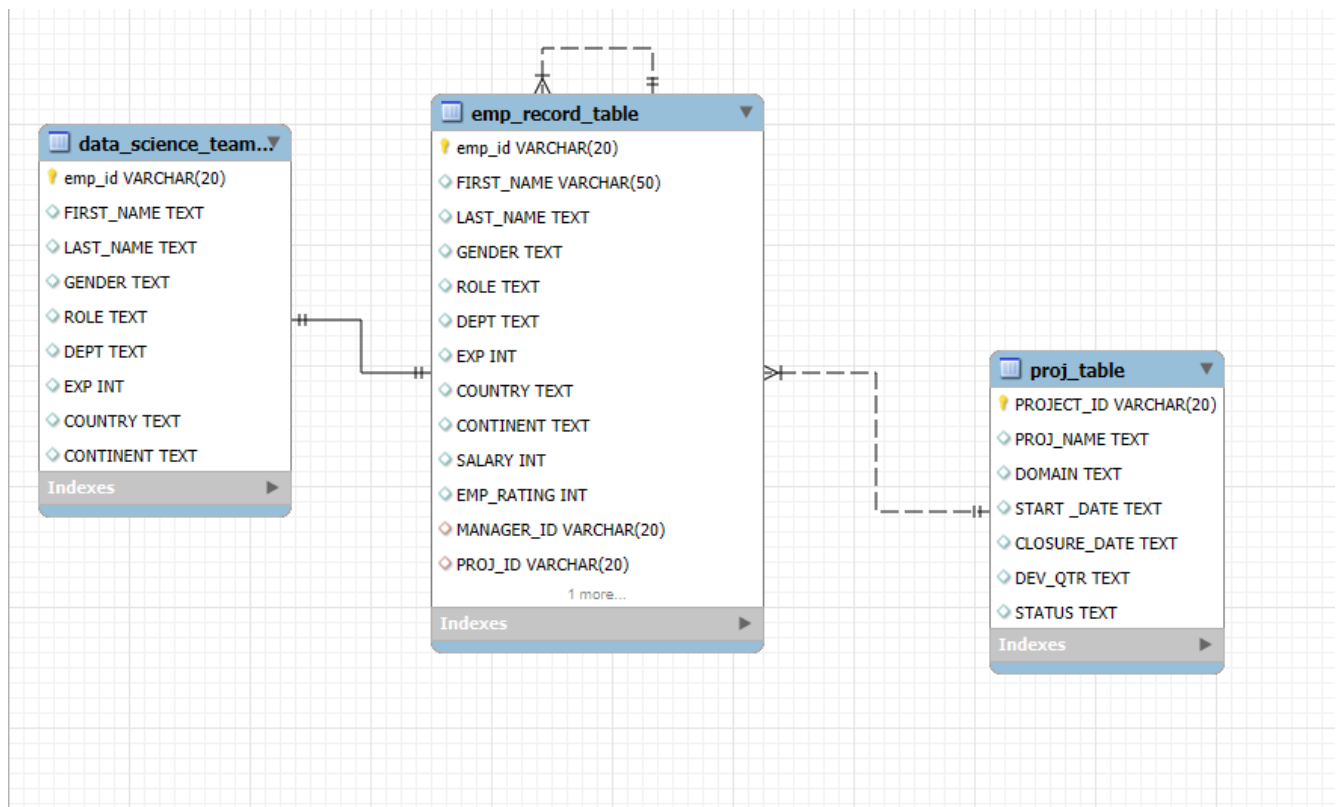


1. Create a database named *employee*, then import **data\_science\_team.csv**, **proj\_table.csv** and **emp\_record\_table.csv** into the **employee** database from the given resources.

```
1 • create database employee;  
2 • use employee;
```

2. Create an ER diagram for the given **employee** database.



**One-to-Many (Self-Referencing):** The relationship between `emp_id` and `manager_id` in the `emp_record_table` is a one-to-many relationship. One employee (`emp_id`) can be the manager of multiple employees (`manager_id`), but each employee can have only one manager.

**Many-to-One:** The relationship between `emp_record_table.proj_id` and `proj_table.project_id` is a many-to-one relationship. Multiple employees (`emp_record_table.proj_id`) can be assigned to the same project (`proj_table.project_id`), but each project can have multiple employees.

**One-to-One:** The relationship between `emp_record_table.emp_id` and `data_science_team.emp_id` is a one-to-one relationship. Each employee in `emp_record_table` may have zero or one corresponding record in `data_science_team`. The relationship is an identifying one-to-one relationship. The primary key of the child table includes (or is derived from) the primary key of the parent table. Row in `data_science_team` cannot exist without a corresponding row in `emp_record_table`, and the child's identity comes from the parent's `emp_id`.

3. Write a query to fetch EMP\_ID, FIRST\_NAME, LAST\_NAME, GENDER, and DEPARTMENT from the employee record table, and make a list of employees and details of their department.

```
3 • SELECT EMP_ID, FIRST_NAME, LAST_NAME, GENDER, DEPT
4 FROM emp_record_table
5 ORDER BY DEPT, FIRST_NAME;
```

| EMP_ID | FIRST_NAME | LAST_NAME | GENDER | DEPT       |
|--------|------------|-----------|--------|------------|
| E001   | Arthur     | Black     | M      | ALL        |
| E532   | Claire     | Brennan   | F      | AUTOMOTIVE |
| E204   | Karene     | Nowak     | F      | AUTOMOTIVE |
| E428   | Pete       | Allen     | M      | AUTOMOTIVE |
| E010   | William    | Butler    | M      | AUTOMOTIVE |
| E103   | Emily      | Grove     | F      | FINANCE    |
| E005   | Eric       | Hoffman   | M      | FINANCE    |
| E403   | Steve      | Hoffman   | M      | FINANCE    |

emp\_record\_table 2 x

4. Write a query to fetch EMP\_ID, FIRST\_NAME, LAST\_NAME, GENDER, DEPARTMENT, and EMP\_RATING if the EMP\_RATING is:
- less than two
  - greater than four
  - between two and four

```

6
7 • SELECT EMP_ID, FIRST_NAME, LAST_NAME, GENDER, DEPT, EMP_RATING
8 FROM emp_record_table
9 WHERE EMP_RATING < 2;
10

```

Result Grid

Filter Rows:

Export:

Wrap Cell Content:

|   | EMP_ID | FIRST_NAME | LAST_NAME | GENDER | DEPT       | EMP_RATING |
|---|--------|------------|-----------|--------|------------|------------|
| ▶ | E057   | Dorothy    | Wilson    | F      | HEALTHCARE | 1          |
|   | E532   | Claire     | Brennan   | F      | AUTOMOTIVE | 1          |
|   | E620   | Katrina    | Allen     | F      | RETAIL     | 1          |

```

11 • SELECT EMP_ID, FIRST_NAME, LAST_NAME, GENDER, DEPT, EMP_RATING
12 FROM emp_record_table
13 WHERE EMP_RATING > 4;

```

Result Grid

Filter Rows:

Export:

Wrap Cell Content:

| EMP_ID | FIRST_NAME | LAST_NAME | GENDER | DEPT       | EMP_RATING |
|--------|------------|-----------|--------|------------|------------|
| E001   | Arthur     | Black     | M      | ALL        | 5          |
| E052   | Dianna     | Wilson    | F      | HEALTHCARE | 5          |
| E083   | Patrick    | Voltz     | M      | HEALTHCARE | 5          |
| E204   | Karene     | Nowak     | F      | AUTOMOTIVE | 5          |

emp\_record\_table 4 ×

```

15 • SELECT EMP_ID, FIRST_NAME, LAST_NAME, GENDER, DEPT, EMP_RATING
16 FROM emp_record_table
17 WHERE (EMP_RATING BETWEEN 2 AND 4)

```

Result Grid

Filter Rows:

Export:

Wrap Cell Content:

|   | EMP_ID | FIRST_NAME | LAST_NAME | GENDER | DEPT       | EMP_RATING |
|---|--------|------------|-----------|--------|------------|------------|
| ▶ | E005   | Eric       | Hoffman   | M      | FINANCE    | 3          |
|   | E010   | William    | Butler    | M      | AUTOMOTIVE | 2          |
|   | E103   | Emily      | Grove     | F      | FINANCE    | 4          |
|   | E245   | Nian       | Zhen      | M      | RETAIL     | 2          |
|   | E260   | Roy        | Collins   | M      | RETAIL     | 3          |
|   | E403   | Steve      | Hoffman   | M      | FINANCE    | 3          |
|   | E428   | Pete       | Allen     | M      | AUTOMOTIVE | 4          |
|   | E478   | David      | Smith     | M      | RETAIL     | 4          |

emp\_record\_table 5 ×

5. Write a query to concatenate the FIRST\_NAME and the LAST\_NAME of employees in the *Finance* department from the employee table and then give the resultant column alias as NAME.

```
20  select CONCAT(FIRST_NAME, ' ', LAST_NAME) AS NAME
21 FROM emp_record_table
22 WHERE DEPT= 'FINANCE';
```

Result Grid  Filter Rows:  Export:  Wrap Cell Content: 

| NAME          |
|---------------|
| Eric Hoffman  |
| Emily Grove   |
| Steve Hoffman |

6. Write a query to list only those employees who have someone reporting to them. Also, show the number of reporters (including the President).

```
SELECT
    m.EMP_ID,
    m.FIRST_NAME,
    COUNT(*) AS REPORTERS
FROM emp_record_table e
JOIN emp_record_table m
    ON e.MANAGER_ID = m.EMP_ID
GROUP BY m.EMP_ID, m.FIRST_NAME;
```

Result Grid  Filter Rows:  Export: 

|   | EMP_ID | FIRST_NAME | REPORTERS |
|---|--------|------------|-----------|
| ▶ | E001   | Arthur     | 5         |
|   | E083   | Patrick    | 3         |
|   | E103   | Emily      | 2         |
|   | E428   | Pete       | 3         |
|   | E583   | Janet      | 3         |
|   | E612   | Tracy      | 2         |

7. Write a query to list down all the employees from the healthcare and finance departments using union. Take data from the employee record table.

```
24 • SELECT EMP_ID, FIRST_NAME, LAST_NAME, GENDER, DEPT
25 FROM emp_record_table
26 WHERE DEPT = 'Healthcare'
27
28 UNION
29
30 SELECT EMP_ID, FIRST_NAME, LAST_NAME, GENDER, DEPT
31 FROM emp_record_table
32 WHERE DEPT = 'Finance';
```

|   | EMP_ID | FIRST_NAME | LAST_NAME | GENDER | DEPT       |
|---|--------|------------|-----------|--------|------------|
| ▶ | E052   | Dianna     | Wilson    | F      | HEALTHCARE |
|   | E057   | Dorothy    | Wilson    | F      | HEALTHCARE |
|   | E083   | Patrick    | Voltz     | M      | HEALTHCARE |
|   | E505   | Chad       | Wilson    | M      | HEALTHCARE |
|   | E005   | Eric       | Hoffman   | M      | FINANCE    |
|   | E103   | Emily      | Grove     | F      | FINANCE    |
|   | E403   | Steve      | Hoffman   | M      | FINANCE    |

8. Write a query to list down employee details such as EMP\_ID, FIRST\_NAME, LAST\_NAME, ROLE, DEPARTMENT, and EMP\_RATING grouped by dept. Also include the respective employee rating along with the max emp rating for the department.

```
--
34 • SELECT
35     EMP_ID,
36     FIRST_NAME,
37     LAST_NAME,
38     ROLE,
39     DEPT,
40     EMP_RATING,
41     MAX(EMP_RATING) OVER (PARTITION BY DEPT) AS MAX_RATING
42 FROM
43     emp_record_table;
```

| Result Grid                       |        |            |           |                          |            |            |            |
|-----------------------------------|--------|------------|-----------|--------------------------|------------|------------|------------|
| Filter Rows: <input type="text"/> |        |            |           |                          |            |            |            |
| Export:  Wrap Cell Content:       |        |            |           |                          |            |            |            |
|                                   | EMP_ID | FIRST_NAME | LAST_NAME | ROLE                     | DEPT       | EMP_RATING | MAX_RATING |
| ▶                                 | E001   | Arthur     | Black     | PRESIDENT                | ALL        | 5          | 5          |
|                                   | E010   | William    | Butler    | LEAD DATA SCIENTIST      | AUTOMOTIVE | 2          | 5          |
|                                   | E204   | Karene     | Nowak     | SENIOR DATA SCIENTIST    | AUTOMOTIVE | 5          | 5          |
|                                   | E428   | Pete       | Allen     | MANAGER                  | AUTOMOTIVE | 4          | 5          |
|                                   | E532   | Claire     | Brennan   | ASSOCIATE DATA SCIENTIST | AUTOMOTIVE | 1          | 5          |
|                                   | E005   | Eric       | Hoffman   | LEAD DATA SCIENTIST      | FINANCE    | 3          | 4          |
|                                   | E103   | Emily      | Grove     | MANAGER                  | FINANCE    | 4          | 4          |
|                                   | E403   | Steve      | Hoffman   | ASSOCIATE DATA SCIENTIST | FINANCE    | 3          | 4          |

Result 11 ×

9. Write a query to calculate the minimum and the maximum salary of the employees in each role. Take data from the employee record table.

```

SELECT
    ROLE,
    MIN(SALARY) AS MIN_SALARY,
    MAX(SALARY) AS MAX_SALARY
FROM
    emp_record_table
GROUP BY
    ROLE;

```

| Result Grid                       |                          |            |            |
|-----------------------------------|--------------------------|------------|------------|
| Filter Rows: <input type="text"/> |                          |            |            |
| Export:  Wrap Cell Content:       |                          |            |            |
|                                   | ROLE                     | MIN_SALARY | MAX_SALARY |
| ▶                                 | PRESIDENT                | 16500      | 16500      |
|                                   | LEAD DATA SCIENTIST      | 8500       | 9000       |
|                                   | SENIOR DATA SCIENTIST    | 5500       | 7700       |
|                                   | MANAGER                  | 8500       | 11000      |
|                                   | ASSOCIATE DATA SCIENTIST | 4000       | 5000       |
|                                   | JUNIOR DATA SCIENTIST    | 2800       | 3000       |

10. Write a query to assign ranks to each employee based on their experience. Take data from the employee record table.

```
SELECT
    EMP_ID,
    FIRST_NAME,
    LAST_NAME,
    ROLE,
    DEPT,
    EXP,
    DENSE_RANK() OVER (ORDER BY EXP DESC) AS EXPERIENCE_RANK
FROM
    emp_record_table;
```

Result Grid

Filter Rows:

Export:

Wrap Cell Content:

|   | EMP_ID | FIRST_NAME | LAST_NAME | ROLE                | DEPT       | EXP | EXPERIENCE_RANK |
|---|--------|------------|-----------|---------------------|------------|-----|-----------------|
| ▶ | E001   | Arthur     | Black     | PRESIDENT           | ALL        | 20  | 1               |
|   | E083   | Patrick    | Voltz     | MANAGER             | HEALTHCARE | 15  | 2               |
|   | E103   | Emily      | Grove     | MANAGER             | FINANCE    | 14  | 3               |
|   | E428   | Pete       | Allen     | MANAGER             | AUTOMOTIVE | 14  | 3               |
|   | E583   | Janet      | Hale      | MANAGER             | RETAIL     | 14  | 3               |
|   | E612   | Tracy      | Norris    | MANAGER             | RETAIL     | 13  | 4               |
|   | E010   | William    | Butler    | LEAD DATA SCIENTIST | AUTOMOTIVE | 12  | 5               |
|   | E005   | Eric       | Hoffman   | LEAD DATA SCIENTIST | FINANCE    | 11  | 6               |

Result 15

×

11. Write a query to create a view that displays employees in various countries whose salary is more than six thousand. Take data from the employee record table.

- ```
CREATE VIEW Employees_Salary_Above_6000 AS
SELECT
    EMP_ID,
    FIRST_NAME,
    LAST_NAME,
    ROLE,
    DEPT,
    COUNTRY,
    SALARY
FROM
    emp_record_table
WHERE
    SALARY > 6000;
```
- ```
SELECT * FROM Employees_Salary_Above_6000;
```

|   | EMP_ID | FIRST_NAME | LAST_NAME | ROLE                  | DEPT       | COUNTRY | SALARY |
|---|--------|------------|-----------|-----------------------|------------|---------|--------|
| ▶ | E001   | Arthur     | Black     | PRESIDENT             | ALL        | USA     | 16500  |
|   | E005   | Eric       | Hoffman   | LEAD DATA SCIENTIST   | FINANCE    | USA     | 8500   |
|   | E010   | William    | Butler    | LEAD DATA SCIENTIST   | AUTOMOTIVE | FRANCE  | 9000   |
|   | E057   | Dorothy    | Wilson    | SENIOR DATA SCIENTIST | HEALTHCARE | USA     | 7700   |
|   | E083   | Patrick    | Voltz     | MANAGER               | HEALTHCARE | USA     | 9500   |
|   | E103   | Emily      | Grove     | MANAGER               | FINANCE    | CANADA  | 10500  |
|   | E204   | Karene     | Nowak     | SENIOR DATA SCIENTIST | AUTOMOTIVE | GERMANY | 7500   |
|   | E245   | Nian       | Zhen      | SENIOR DATA SCIENTIST | RETAIL     | CHINA   | 6500   |

Employees\_Salary\_Above\_6000... x

12. Write a nested query to find employees with experience of more than ten years. Take data from the employee record table.

```
SELECT EMP_ID, FIRST_NAME, LAST_NAME, ROLE, DEPT, EXP
FROM emp_record_table
WHERE EXP = ANY (
    SELECT EXP
    FROM emp_record_table
    WHERE EXP > 10
);
```

| Result Grid   Filter Rows: <input type="text"/>   Export:    Wrap Cell Content:  |        |            |           |                     |            |     |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------|------------|-----------|---------------------|------------|-----|
|                                                                                                                                                                                                                                                                                                                                                                                                                                | EMP_ID | FIRST_NAME | LAST_NAME | ROLE                | DEPT       | EXP |
| ▶                                                                                                                                                                                                                                                                                                                                                                                                                              | E001   | Arthur     | Black     | PRESIDENT           | ALL        | 20  |
|                                                                                                                                                                                                                                                                                                                                                                                                                                | E005   | Eric       | Hoffman   | LEAD DATA SCIENTIST | FINANCE    | 11  |
|                                                                                                                                                                                                                                                                                                                                                                                                                                | E010   | William    | Butler    | LEAD DATA SCIENTIST | AUTOMOTIVE | 12  |
|                                                                                                                                                                                                                                                                                                                                                                                                                                | E083   | Patrick    | Voltz     | MANAGER             | HEALTHCARE | 15  |
|                                                                                                                                                                                                                                                                                                                                                                                                                                | E103   | Emily      | Grove     | MANAGER             | FINANCE    | 14  |
|                                                                                                                                                                                                                                                                                                                                                                                                                                | E428   | Pete       | Allen     | MANAGER             | AUTOMOTIVE | 14  |
|                                                                                                                                                                                                                                                                                                                                                                                                                                | E583   | Janet      | Hale      | MANAGER             | RETAIL     | 14  |
|                                                                                                                                                                                                                                                                                                                                                                                                                                | E612   | Tracy      | Norris    | MANAGER             | RETAIL     | 13  |

13. Write a query to create a stored procedure to retrieve the details of the employees whose experience is more than three years. Take data from the employee record table.

DELIMITER //

```
• CREATE PROCEDURE GetEmployeesWithExperience()  
  BEGIN  
    SELECT EMP_ID, FIRST_NAME, LAST_NAME, ROLE, DEPT, EXP, SALARY  
    FROM emp_record_table  
    WHERE EXP > 3;  
  END //
```

DELIMITER ;

```
• CALL GetEmployeesWithExperience();
```

Result Grid

Filter Rows:

Export:

Wrap Cell Content:

|   | EMP_ID | FIRST_NAME | LAST_NAME | ROLE                  | DEPT       | EXP | SALARY |
|---|--------|------------|-----------|-----------------------|------------|-----|--------|
| ▶ | E001   | Arthur     | Black     | PRESIDENT             | ALL        | 20  | 16500  |
|   | E005   | Eric       | Hoffman   | LEAD DATA SCIENTIST   | FINANCE    | 11  | 8500   |
|   | E010   | William    | Butler    | LEAD DATA SCIENTIST   | AUTOMOTIVE | 12  | 9000   |
|   | E052   | Dianna     | Wilson    | SENIOR DATA SCIENTIST | HEALTHCARE | 6   | 5500   |
|   | E057   | Dorothy    | Wilson    | SENIOR DATA SCIENTIST | HEALTHCARE | 9   | 7700   |
|   | E083   | Patrick    | Voltz     | MANAGER               | HEALTHCARE | 15  | 9500   |
|   | E103   | Emily      | Grove     | MANAGER               | FINANCE    | 14  | 10500  |
|   | E204   | Karene     | Nowak     | SENIOR DATA SCIENTIST | AUTOMOTIVE | 8   | 7500   |

14. Write a query using stored functions in the project table to check whether the job profile assigned to each employee in the data science team matches the organization's set standard.

The standard being:

For an employee with experience less than or equal to 2 years assign 'JUNIOR DATA SCIENTIST',

For an employee with the experience of 2 to 5 years assign 'ASSOCIATE DATA SCIENTIST',

For an employee with the experience of 5 to 10 years assign 'SENIOR DATA SCIENTIST',

For an employee with the experience of 10 to 12 years assign 'LEAD DATA SCIENTIST',

For an employee with the experience of 12 to 16 years assign 'MANAGER'.

```
DELIMITER //
```

```
CREATE FUNCTION employee_job_profile(EXP INT)
```

```
RETURNS VARCHAR(50)
```

```
DETERMINISTIC
```

```
BEGIN
```

```
    DECLARE standard_role VARCHAR(50);
```

```
    IF EXP <= 2 THEN
```

```
        SET standard_role = 'JUNIOR DATA SCIENTIST';
```

```
    ELSEIF EXP <=5 THEN
```

```
        SET standard_role = 'ASSOCIATE DATA SCIENTIST';
```

```
    ELSEIF EXP <=10 THEN
```

```
        SET standard_role = 'SENIOR DATA SCIENTIST';
```

```
    ELSEIF EXP <=12 THEN
```

```
        SET standard_role = 'LEAD DATA SCIENTIST';
```

```
    ELSEIF EXP <=16 THEN
```

```
        SET standard_role = 'MANAGER';
```

```
    END IF;
```

```
    RETURN standard_role;
```

```
END //
```

```
DELIMITER ;
```

```

SELECT
    e.emp_id,
    e.first_name,
    e.last_name,
    e.role AS Assigned_Role,
    employee_job_profile(e.EXP) AS Expected_Role,
    CASE
        WHEN e.role = employee_job_profile(e.EXP) THEN 'MATCHES'
        ELSE 'DOES NOT MATCH'
    END AS Role_Match_Status
FROM
    data_science_team e;

```

|   | emp_id | first_name | last_name | Assigned_Role            | Expected_Role            | Role_Match_Status |
|---|--------|------------|-----------|--------------------------|--------------------------|-------------------|
| ▶ | E005   | Eric       | Hoffman   | LEAD DATA SCIENTIST      | LEAD DATA SCIENTIST      | MATCHES           |
|   | E010   | William    | Butler    | LEAD DATA SCIENTIST      | LEAD DATA SCIENTIST      | MATCHES           |
|   | E052   | Dianna     | Wilson    | SENIOR DATA SCIENTIST    | SENIOR DATA SCIENTIST    | MATCHES           |
|   | E057   | Dorothy    | Wilson    | SENIOR DATA SCIENTIST    | SENIOR DATA SCIENTIST    | MATCHES           |
|   | E204   | Karene     | Nowak     | SENIOR DATA SCIENTIST    | SENIOR DATA SCIENTIST    | MATCHES           |
|   | E245   | Nian       | Zhen      | SENIOR DATA SCIENTIST    | SENIOR DATA SCIENTIST    | MATCHES           |
|   | E260   | Roy        | Collins   | SENIOR DATA SCIENTIST    | SENIOR DATA SCIENTIST    | MATCHES           |
|   | E403   | Steve      | Hoffman   | ASSOCIATE DATA SCIENTIST | ASSOCIATE DATA SCIENTIST | MATCHES           |

Result 24 x

15. Create an index to improve the cost and performance of the query to find the employee whose FIRST\_NAME is 'Eric' in the employee table after checking the execution plan.

-- Check execution plan before creating index:

```

explain select EMP_ID, FIRST_NAME, LAST_NAME
from emp_record_table
where FIRST_NAME = 'Eric';

```

|   | id | select_type | table            | partitions | type | possible_keys | key  | key_len | ref  | rows | filtered | Extra       |
|---|----|-------------|------------------|------------|------|---------------|------|---------|------|------|----------|-------------|
| ▶ | 1  | SIMPLE      | emp_record_table | NULL       | ALL  | NULL          | NULL | NULL    | NULL | 19   | 10.00    | Using where |

-- Create index

```

create index idx_first_name on emp_record_table(FIRST_NAME);

```

-- Check execution plan after creating index

```

explain select EMP_ID, FIRST_NAME, LAST_NAME
from emp_record_table
where FIRST_NAME = 'Eric';

```

|   | id | select_type | table            | partitions | type | possible_keys  | key            | key_len | ref   | rows | filtered | Extra |
|---|----|-------------|------------------|------------|------|----------------|----------------|---------|-------|------|----------|-------|
| ▶ | 1  | SIMPLE      | emp_record_table | HULL       | ref  | idx_first_name | idx_first_name | 203     | const | 1    | 100.00   | HULL  |

16. Write a query to calculate bonus for all employees, based on their ratings and salaries (Use the formula: 5% of salary \* employee rating).

```
6 • select EMP_ID, FIRST_NAME, LAST_NAME,
7      ROLE, DEPT, EXP, SALARY, EMP_RATING,
8      (0.05*SALARY) * EMP_RATING AS Bonus
9      from emp_record_table;
```

| Result Grid     Filter Rows: <input type="text"/>   Export:    Wrap Cell Content: |            |           |                       |            |     |        |            |         |  |
|-----------------------------------------------------------------------------------|------------|-----------|-----------------------|------------|-----|--------|------------|---------|--|
| ID                                                                                | FIRST_NAME | LAST_NAME | ROLE                  | DEPT       | EXP | SALARY | EMP_RATING | Bonus   |  |
| ▶                                                                                 | Arthur     | Black     | PRESIDENT             | ALL        | 20  | 16500  | 5          | 4125.00 |  |
|                                                                                   | Eric       | Hoffman   | LEAD DATA SCIENTIST   | FINANCE    | 11  | 8500   | 3          | 1275.00 |  |
|                                                                                   | William    | Butler    | LEAD DATA SCIENTIST   | AUTOMOTIVE | 12  | 9000   | 2          | 900.00  |  |
|                                                                                   | Dianna     | Wilson    | SENIOR DATA SCIENTIST | HEALTHCARE | 6   | 5500   | 5          | 1375.00 |  |
|                                                                                   | Dorothy    | Wilson    | SENIOR DATA SCIENTIST | HEALTHCARE | 9   | 7700   | 1          | 385.00  |  |
|                                                                                   | Patrick    | Voltz     | MANAGER               | HEALTHCARE | 15  | 9500   | 5          | 2375.00 |  |
|                                                                                   | Emily      | Grove     | MANAGER               | FINANCE    | 14  | 10500  | 4          | 2100.00 |  |

17. Write a query to calculate the average salary distribution based on the continent and country. Take data from the employee record table.

```
• SELECT
    continent,
    country,
    AVG(salary) AS avg_salary
FROM
    emp_record_table
GROUP BY
    continent,
    country
ORDER BY
    continent,
    country;
```

| Result Grid     Filter Rows: <input type="text"/> |               |          |            |
|---------------------------------------------------|---------------|----------|------------|
|                                                   | continent     | country  | avg_salary |
| ▶                                                 | ASIA          | CHINA    | 6500.0000  |
|                                                   | ASIA          | INDIA    | 6166.6667  |
|                                                   | EUROPE        | FRANCE   | 9000.0000  |
|                                                   | EUROPE        | GERMANY  | 7600.0000  |
|                                                   | NORTH AMERICA | CANADA   | 7000.0000  |
|                                                   | NORTH AMERICA | USA      | 9440.0000  |
|                                                   | SOUTH AMERICA | COLOMBIA | 5600.0000  |







